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The structure of dynamic correlations in multivariate stochastic volatility models[☆]

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ABSTRACT

This paper proposes two types of stochastic correlation structures for Multivariate Stochastic Volatility (MSV) models, namely the constant correlation (CC) MSV and dynamic correlation (DC) MSV models, from which the stochastic covariance structures can easily be obtained. Both structures can be used for purposes of determining optimal portfolio and risk management strategies through the use of correlation matrices, and for calculating Value-at-Risk (VaR) forecasts and optimal capital charges under the Basel Accord through the use of covariance matrices. A technique is developed to estimate the DC MSV model using the Markov Chain Monte Carlo (MCMC) procedure, and simulated data show that the estimation method works well. Various multivariate conditional volatility and MSV models are compared via simulation, including an evaluation of alternative VaR estimators. The DC MSV model is also estimated using three sets of empirical data, namely Nikkei 225 Index, Hang Seng Index and Straits Times Index returns, and significant dynamic correlations are found. The Dynamic Conditional Correlation (DCC) model is also estimated, and is found to be far less sensitive to the covariation in the shocks to the indexes. The correlation process for the DCC model also appears to have a unit root, and hence constant conditional correlations in the long run. In contrast, the estimates arising from the DC MSV model indicate that the dynamic correlation process is stationary.

1. Introduction

Static and dynamic covariance and correlation structures are used routinely for optimal portfolio choice, risk management, obtaining Value-at-Risk (VaR) forecasts, and determining optimal capital charges under the Basel Accord. Although the conditional volatility literature has examined the theoretical development of alternative dynamic covariance and correlation structures, this issue does not yet seem to have been examined in detail in the Multivariate Stochastic Volatility (MSV) literature.

For multivariate GARCH models, the most general expression is called the 'vec' model (see Engle and Kroner (1995)). The vec model parameterizes the vector of the conditional covariance matrix of the returns vector, which is determined by its lags and the vector of outer products of the lagged returns vector. A serious issue with the vec model is that it has many parameters to be estimated, and will not guarantee positive definiteness of the conditional covariance matrix without further restrictions. Bollerslev et al. (1988) and Ding and Engle (2001) suggested the diagonal GARCH model, which restricts the off-diagonal elements of the parameter matrices to be zero, and also reduces the number of parameters drastically in computing the conditional covariance matrix. Engle and Kroner (1995) proposed the Baba, Engle, Kraft and Kroner (BEKK) specification that guarantees the positive definiteness of the conditional covariance matrix, which is essential for obtaining sensible VaR forecasts.

In the context of modelling conditional correlations rather than conditional covariances, Bollerslev (1990) proposed the Constant Conditional Correlation (CCC) model, where the time-varying covariances are proportional to the conditional standard deviation derived from univariate GARCH processes. This specification also guarantees the positive definiteness of the conditional covariance matrix. Ling and McAleer (2003) develop the asymptotic theory for several constant correlation vector ARMA–GARCH models. As an extension of the CCC model, Engle (2002) suggested the Dynamic Conditional Correlation (DCC) model, which allows the conditional correlation matrix to vary parsimoniously over time.

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Table 1
Constant and dynamic correlation multivariate GARCH models.

Model	Specification	Number of parameters	Type
CCC: Bollerslev (1990)	$P_t = P$, $\omega_{ii,t} = w_i + \alpha_i \varepsilon^2_{ii,t-1} + \beta_i \omega_{ii,t-1}$	$\frac{m^2+5m}{2}$	C
DCC: Engle (2002)	$P_t = Q_t^{*-1} Q_t Q_t^{*-1}$, $Q_t = (1-\gamma-\delta) S + \gamma z_{t-1} z'_{t-1} + \delta Q_{t-1}$, $Q_t^* = \text{diag}\left\{\sqrt{q_{11,t}}, \ldots \sqrt{q_{mm,t}}\right\}$, $z_t = D_t^{-1} \varepsilon_t$, $\omega_{ii,t} = w_i + \alpha_i \varepsilon^2_{ii,t-1} + \beta_i \omega_{ii,t-1}$	$\frac{m^2+5m}{2} + 2$	D
BEKK: Engle and Kroner (1995)	$\Omega_t = C'C + A\varepsilon_{t-1}\varepsilon'_{t-1}A' + B\Omega_{t-1}B'$	$\frac{5m^2+m}{2}$	D
Diagonal GARCH: Ding and Engle (2001)	$\Omega_t = C + A \circ \varepsilon_{t-1}\varepsilon'_{t-1} + B \circ \Omega_{t-1}$	$\frac{3m^2+3m}{2}$	D

In the column named 'Type', 'D' denotes dynamic conditional correlation models and 'C' denotes the constant conditional correlation model.

The development of dynamic correlation and covariance models has proceeded at a faster pace in the conditional volatility literature than in its stochastic volatility counterpart. Two reasons for this would seem to be the development of parsimonious multivariate dynamic conditional correlation models and their relative ease in estimation. McAleer (2005) provides a comprehensive comparison of a wide range of univariate and multivariate, conditional and stochastic, financial volatility models. Asai et al. (2006) discuss recent theoretical developments in the MSV literature.

Recently, Yu and Meyer (2006) developed the time-varying correlation model for the bivariate SV model, based on the Fisher transformation, as suggested by Tsay (2002) in a bivariate GARCH framework. Yu and Meyer (2006) also compared the empirical performance of nine alternative MSV models for a bivariate exchange rate series and found that MSV models that allow for time-varying correlations generally fit the data better. An obvious drawback of their analysis is the difficulty in generalizing their dynamic correlation model to a higher dimension. Yu and Meyer (2006, p. 366), note that "it is not easy to generalize the model into higher dimensional situations". The dynamic correlation MSV models that are developed in this paper are not restricted to be bivariate.

As a contribution to the development of parsimonious dynamic correlation MSV models that can be estimated with relative ease, Section 2 proposes two types of stochastic correlation structures for MSV models, namely the constant correlation (CC) MSV and dynamic correlation (DC) MSV models. The dynamic stochastic covariance matrices may be obtained easily from the dynamic stochastic correlation matrices. Alternative DC MSV models are developed. Both structures can be used for purposes of determining optimal portfolio and risk management strategies through the use of dynamic correlations, and for calculating Value-at-Risk (VaR) forecasts and optimal capital charges under the Basel Accord through the use of dynamic covariances. A technique is developed in Section 3 for estimating the DC MSV model using the Markov Chain Monte Carlo (MCMC) procedure. The properties of the estimation method are examined using simulated data, and various multivariate conditional volatility and MSV models are compared via simulation, including an evaluation of alternative VaR estimators. Section 4 provides an empirical example in which the model is estimated using three sets of empirical data. Some concluding remarks are given in Section 5.

2. Dynamic correlation models

In this section, the following definitions are used. Let ε_t be an m-dimensional stochastic vector. The operator vecd(.) creates a vector from the diagonal elements of a matrix. The operator $\circ$ denotes the Hadamard (or element-by-element) product. Let exp(.) denote the element-by-element exponential operator, and $\text{diag}\{x\} = \text{diag}\{x_1, \ldots, x_m\}$ denote the $m \times m$ diagonal matrix, with diagonal elements given by $x = (x_1, \ldots, x_m)'$. For any $m \times m$ matrix A, the matrix exponential transformation is defined by the power series expansion:

$$\text{Exp}(A) = \sum_{s=0}^{\infty} (1/s!) A^S,$$

where A^0 reduces to the $m \times m$ identity matrix and A^s denotes the standard matrix multiplication of A s times. Thus, in general, the elements of Exp(A) do not typically exponentiate the elements of A.

2.1. Multivariate conditional volatility models

In the framework of the conditional volatility model, it is assumed that $E(\varepsilon_t \mid \Im_{t-1}) = 0$ and $E\left(\varepsilon_t \varepsilon'_t \mid \Im_{t-1}\right) = \Omega_t$, where $\Im_t$ is an information set up to period t. Thus, $\Omega_t = \left\{\omega_{ij,t}\right\}$ is the covariance matrix of ε_t conditional on past information. Let $D_t = \text{diag}\left\{\sqrt{\omega_{11,t}}, \ldots \sqrt{\omega_{mm,t}}\right\}$, so that the dynamic conditional correlation matrix, P_t, is defined by

$$P_t = D_t^{-1} \Omega_t D_t^{-1}. \quad (1)$$

The dynamic conditional covariance matrix, Ω_t, can be obtained from (1) by pre- and post-multiplication of both sides by the diagonal matrix to yield $\Omega_t = D_t P_t D_t$.

Table 1 shows the constant conditional correlation model and three dynamic conditional correlation models, namely, the CCC model of Bollerslev (1990), the DCC model of Engle (2002), the BEKK model of Engle and Kroner (1995), and the diagonal GARCH model of Ding and Engle (2001). The first two models are based on the specification of the conditional correlation matrix, P_t, while the remaining two models are based on the conditional covariance matrix, Ω_t. For the BEKK and diagonal GARCH models, the conditional correlation matrix defined by is dynamic. The CCC and DCC models are parsimonious, whereas the BEKK and diagonal GARCH models are not. The latter two models can be made more parsimonious by the imposition of suitable parametric restrictions. In multivariate conditional volatility models, the number of parameters increases to the order of m^2. When $m = 5$ (10), the numbers of parameters in the CCC and DCC models are 25 (75) and 27 (77), respectively, while those in the BEKK and diagonal GARCH models are 65 (255) and 45 (165),respectively.

Given the above, the primary features of the DCC model are that (i) each ε_{it} follows the univariate GARCH model, as in the estimation of the CCC model, which essentially follows a multiple univariate structure, and (ii) dynamic conditional correlations can be obtained through the addition of only two parameters to the CCC model. Thus, the DCC model is parsimonious in capturing dynamic correlations and covariances.

2.2. Multivariate stochastic volatility models

For MSV models, it is assumed that $E(\varepsilon_t \mid \Omega_t) = 0$ and $E\left(\varepsilon_t \varepsilon'_t \mid \Omega_t\right) = \Omega_t$, where the covariance matrix, Ω_t, is stochastic and symmetric positive definite. The first MSV model, which will

Table 2
Constant and dynamic correlation MSV models.

Model	Specification	Number of parameters	Type
Basic MSV: Harvey et al. (1994)	Eq. (2), $\nu_t \sim N(0, P)$, $\eta_t \sim N(0, \Sigma_\eta)$	$m^2 + 2m$	C
CC MSV (special case of Basic MSV): This paper	Eq. (2), $\nu_t \sim N(0, P)$, $\eta_t \sim N\left(0, \text{diag}\left\{\sigma_1^2, \ldots, \sigma_m^2\right\}\right)$	$\frac{m^2+5m}{2}$	C
WAR: Gourieroux et al. (2005), Gourieroux (2006)	$\Omega_t = \sum_{i=1}^m x_{it}x_{it}'$, $x_{it} = Ax_{i,t-1} + \varepsilon_{it}$, $\varepsilon_{it} \sim N(0, \Sigma)$	$\frac{3m^2+m}{2}$	D
Matrix Exponential MSV: Asai et al. (2006)	$\Sigma_t = \text{Exp}(A_t)$, $\alpha_t = \text{vech}(A_t)$, $\alpha_{t+1} = c + \psi \circ \alpha_t + u_t$, $u_t \sim N\left(0, \text{diag}\left\{\sigma_1^2, \ldots, \sigma_{m(m+1)/2}^2\right\}\right)$	$\frac{3m^2+3m}{2}$	D
Wishart Inverse Covariance (WIC): This paper	$\Omega_t^{-1} \mid k, S_{t-1} \sim W_m(k, S_{t-1})$, $S_t = \frac{1}{k}\Omega_t^{-d/2}A\Omega_t^{-d/2}$, $k \sim$ scalar, $A \sim$ p.d.	$\frac{m^2+m}{2} + 2$	D
DCMSV1: This paper	Eq. (4) based on CC MSV except for P_t	$m^2 + 4m + 2$	D
DCMSV2: This paper	Eq. (5) based on CC MSV except for P_t	$\frac{m^2+7m}{2} + 2$	D

In the column named 'Type', 'D' denotes dynamic correlation MSV models and 'C' denotes constant correlation MSV models.

be called the basic MSV model, was proposed by Harvey et al. (1994) and is given as follows:

$$\begin{aligned} \varepsilon_t &= D_t \nu_t, \\ D_t &= \text{diag}\left\{\exp\left(0.5h_t\right)\right\}, \\ h_{t+1} &= \mu + \phi \circ h_t + \eta_t \end{aligned} \tag{2}$$

where the shocks to returns and volatility, ν_t and η_t, respectively, are independent processes and are distributed as $\nu_t \sim N(0, P)$ and $\eta_t \sim N\left(0, \Sigma_\eta\right)$. Given this MSV specification, it follows that $\Omega_t = D_t P_t D_t$, as given in, where the correlation matrix may be constant or dynamic. When $m = 5\,(10)$, the number of parameters is 35 (120).

It should be noted that the basic MSV model of Harvey et al. (1994) is not the MSV counterpart of the CCC model, as the former incorporates the simultaneous correlation between volatilities through the covariance matrix of η_t. The MSV counterpart of the CCC model is given by with $\eta_t \sim N\left(0, \text{diag}\left\{\sigma_{\eta,1}^2, \ldots, \sigma_{\eta,m}^2\right\}\right)$, which will be referred to as the 'Constant Correlation MSV' (CC MSV) model. The number of parameters in the CC MSV model is the same as that of the CCC model which, for $m = 5(10)$, is 25(75).

As in the case of multivariate GARCH models, there are two ways of developing dynamic correlation models, one based on the specification of P_t and the other based on Ω_t. For the latter, two MSV models are available in the literature, namely the Wishart AutoRegressive (WAR) model of Gourieroux et al. (2004) and Gourieroux (2006), and the matrix exponential MSV model of Asai et al. (2006). The Wishart distribution is a generalization of a chi-squared distribution in a multivariate framework. Table 2 shows the specifications of the alternative constant and dynamic correlation models. It should be noted that the matrix exponential model is restricted to have a diagonal covariance matrix for the volatility innovations. When $m = 5(10)$, the numbers of parameters for the WAR and matrix exponential models are 40 (155) and 45 (165), respectively.

In addition to the above models, we consider an alternative Wishart model, namely the 'Wishart Inverse Covariance' (WIC), which is also given in Table 2. A related model was proposed by Philipov and Glickman (2006a,b), but their specification uses $S_t = (1/k)\left(A^{1/2}\right)\Omega_t^{-d}\left(A^{1/2}\right)'$ and a specific MCMC estimation method was proposed. As the WIC model proposed in this paper is different from the factor MSV models of Philipov and Glickman (2006a,b), the sampling scheme estimator will also be different. Furthermore, Philipov and Glickman (2006a,b) restricted the degrees of freedom parameter, k, of the Wishart distribution to be a positive integer, with $k \geq m$, but this assumption can be relaxed to allow the parameter k to take any real value.

The reason for specifying Ω_t^{-1} instead of Ω_t arises from Bayesian analysis of multiple equations models, in which the prior distribution of the inverse of the covariance matrix of the error vector is assumed to follow the Wishart distribution, so that the posterior of the inverse of the covariance matrix also follows the Wishart distribution. In this context, the covariance matrix of the returns vector follows the Wishart distribution conditional on the past information. When $m = 5(10)$, the number of parameters in the parsimonious WIC model is 17 (57), which is more parsimonious than the CCC and DCC models.

We now turn to the dynamic correlation (DC) MSV models based on P_t. The specifications are based on the CC MSV model and the standardization suggested by Engle (2002) in the context of dynamic conditional correlation models, that is,

$$\begin{aligned} P_t &= Q_t^{*-1} Q_t Q_t^{*-1}, \\ Q_t^* &= \left(\text{diag}\left\{\text{vecd}\left(Q_t\right)\right\}\right)^{1/2}, \end{aligned} \tag{3}$$

where Q_t is a sequence of positive definite matrices. The standardization is required to obtain a dynamic stochastic correlation matrix. Although it is possible to consider alternative DC MSV models based on Harvey et al. (1994), we prefer the more convenient structure that each ε_{it} is assumed to follow a univariate SV model. In what follows, we consider two types of parsimonious DC MSV models that impose symmetric positive definiteness on the covariance matrix.

The first DC MSV model, which will be labelled as DCMSV1, is given as follows:

DCMSV1 Model

$$\begin{aligned} Q_{t+1} &= (1-\psi)\bar{Q} + \psi Q_t + \Xi_t, \\ \Xi_t &\sim W_m(k, \Lambda), \end{aligned} \tag{4}$$

in which the MSV shocks are assumed to follow a Wishart process, where $W_m(k, \Lambda)$ denotes a Wishart distribution with degrees of freedom parameter, k, and scale matrix, Λ, respectively. The DCMSV1 model guarantees the symmetric positive definiteness of P_t under the assumption that $\bar{Q}$ is positive definite and $|\psi| < 1$. It is possible to consider a generalization of the model as $Q_{t+1} = (\iota\iota' - \Psi) \circ \bar{Q} + \Psi \circ Q_t + \Xi_t$, which corresponds to a generalization of the DCC model. When $m = 5(10)$, the number of parameters in DCMSV1 is 47 (142).

An alternative DC MSV model, which will be labeled as DCMSV2, is given as follows:

DCMSV2 Model

$$\begin{aligned} Q_{t+1}^{-1} \mid k, S_t &\sim W_m(k, S_t), \\ S_t &= \frac{1}{k} Q_t^{-d/2} A Q_t^{-d/2}, \end{aligned} \tag{5}$$

where k and S_t are the degrees of freedom and the time-dependent scale parameter of the Wishart distribution, respectively, A is a positive definite symmetric parameter matrix, d is a scalar parameter, and $Q_t^{-d/2}$ is defined by using a singular value decomposition. The quadratic expression, together with $k \geq m$, ensures that the covariance matrices are symmetric positive definite. The process of Q_t in the DCMSV2 model has the same structure as in the WIC model. For convenience of identification of the parameters, it is assumed that $Q_0 = I_m$. When $m = 5(10)$, the number of parameters for the DCMSV2 model is 32 (87).

Instead of S_t, as defined above, it is also possible to use

$$S_t = \frac{1}{k}\left[A \circ \left(Q_t^{-1}\right)^d\right] \quad \text{or} \quad S_t = \frac{1}{k}\left(A^{1/2}\right)\left(Q_t^{-1}\right)^d\left(A^{1/2}\right)',$$

for which the parameterization may be based on the diagonal GARCH or BEKK specifications.

Let $\Xi_t \sim W_m(k, I_m)$, so that from it follows that:

$$Q_{t+1}^{-1} = \frac{1}{k} Q_t^{-d/2} A^{1/2} \Xi_t A^{1/2} Q_t^{-d/2}.$$

Taking logarithms of the determinants of both sides yields the following:

$$\ln\left|Q_{t+1}^{-1}\right| = \ln\left|k^{-1}A\right| + d\ln\left|Q_t^{-1}\right| + \ln\left|\Xi_t\right|,$$

so that the condition for stationarity is given by $|d| < 1$.

As compared with the CC MSV model, the DCMSV1 and DCMSV2 models need $\left(m^2 + 3m + 4\right)/2$ and $m + 2$ additional parameters, respectively, with the fewer additional parameters being an obvious merit of the more parsimonious DCMSV2. Second, the distribution of Q_t is unknown for the DCMSV1 model, while Q_t^{-1} follows the Wishart distribution for the DCMSV2 model. Third, the DCMSV2 model reduces to the CC MSV model if and only if $d = 0$, while such a condition is vague for the DCMSV1 model. Thus, dynamics will be present when d is greater than zero.

For the reasons given above, the DCMSV2 model will be considered in the remainder of the paper. For convenience, the DCMSV2 model will be referred to as the DC MSV model. In the next section, we will develop an MCMC technique to estimate this particular DC MSV model.

3. MCMC estimation

3.1. General scheme

One of the most popular approaches for inference of MSV models is the Markov chain Monte Carlo (MCMC) method, including Gibbs sampling and the Metropolis–Hastings algorithm (see, for example, Jacquier et al. (1994), Kim et al. (1998), and Asai et al. (2006)). In this section, we propose an MCMC technique to estimate the DC MSV model. One advantage of the MCMC method as compared with the simulated likelihood approaches is that it is possible to obtain estimates of the parameters and unobservable components simultaneously.

As a Bayesian approach, the idea behind MCMC methods is to produce variates from a given multivariate density (namely, the posterior density in Bayesian applications) by repeatedly sampling a Markov chain whose invariant distribution is the target density of interest. The MCMC method focuses on the density, $\pi(\theta, h, Q \mid y)$, instead of the usual posterior density, $\pi(\theta \mid y)$, as the latter requires computation of the likelihood function. As a result, the parameter space is augmented by including all the latent variables. A cyclic chain of the MCMC procedure is produced by sampling from each conditional distribution and updating each variate. After the Markov chain converges to the target distribution, these draws can be treated as the samples from marginal posterior densities. These draws can be used as the basis for drawing inferences by appealing to suitable ergodic theorems for Markov chains. For example, posterior moments and marginal densities can be estimated (or simulated consistently) by averaging the relevant function of interest over the sampled variates. The posterior mean of θ is estimated simply as the sample mean of the simulated θ values. These estimates can be made arbitrarily accurate by increasing the simulation sample size.

It is well known that estimating CC MSV models via the MCMC approach is computationally demanding. In addition to the unknown parameters and volatilities in the CC MSV model, the new DC MSV model developed here has an unobservable sequence of $m \times m$ matrices, $\{Q_t\}$. For practical purposes, it is necessary to develop a procedure in which feasibility, speed, numerical accuracy and stability are well balanced. Consequently, we propose the following two stage procedure:

(i) For the i-th series, $\{\varepsilon_{it}\}$, estimate the parameters $\left(\mu_i, \phi_i, \sigma_{\eta,i}\right)$ and volatilities $(h_{i1}, \ldots h_{iT})$ via the MCMC method, such as Jacquier et al. (1994), Kim et al. (1998), and Asai (2005). Obtain the standardized series $z_{it} = \varepsilon_{it} W_{it}$, where W_{it} are MCMC estimates of the inverse of volatility obtained by averaging MCMC draws as $W_{it} = (1/R)\sum_{r=1}^{R}\exp(-h_{it}^{(r)})$.
(ii) Based on the standardized vector, $z_t = (z_{1t}, \ldots, z_{mt})'$, estimate the parameters (A, d, k) and $\{Q_t\}$ via the MCMC method.

By using the first stage, we can save computational time in the sense that estimating individual equations m times in multivariate volatility models is faster than estimating a system of m equations. As each process, ε_{it}, follows a univariate SV model, the estimates of the volatilities based on the first stage is valid. Based on the estimates of $\{h_t\}$ and $\{Q_t\}$, we can obtain MCMC estimates of $\{\Omega_t\}$. It should be noted that we can conduct the above MCMC estimation simultaneously. For instance, the empirical results in Section 4 are based on the above procedure, but the results were unchanged when we estimate the DC MSV model as a system.

The structure of the MCMC sampler for the second stage is as follows:

(1) Initialize (A, d, k) and $\{Q_t\}$.
(2) Sample Q_t from $Q_t \mid Q_{\backslash t}, A, d, k, z$ for $t = 1, \ldots, T$.
(3) Sample $A \mid Q, d, k, z$.
(4) Sample $d \mid Q, A, k, z$.
(5) Sample $k \mid Q, A, d, z$.
(6) Go to step (2).

Cycling through steps (2)–(6) is a complete sweep of the MCMC sampler. In the following subsection, we will explain the prior distribution and the likelihood. The conditional distribution for each step is given in the Appendix.

It should be noted that the two-step procedure given above is unusual for Bayesian analysis, and hence we should mention the case of joint estimation. First, MCMC estimation of all the parameters and the latent process is time consuming if the number of variables, m, is large, as in the case of estimating Value-at-Risk (VaR) of a portfolio. Second, the estimates of the volatilities in the first step are very close to those obtained from joint estimation, which parallels estimation of the CCC and DCC conditional volatility models. For these reasons, we choose a convenient method for estimating time-varying covariance matrices, even for dynamic models with large m.

3.2. Implementation issues

For the unknown parameters for each SV process, we employ the MCMC method of Kim et al. (1998), as the approach is well cited in the literature. Following Kim et al. (1998), we work with the prior distributions $\mu_i \sim N\left(\mu_0, \sigma_{\mu 0}^2\right)$, $\left(\phi_i + 1\right)/2 \sim Beta\,(c_1, c_2)$, and $\sigma_{\eta,i} \sim IG\left(k_\eta, S_\eta\right)$, where $Beta\,(c_1, c_2)$ and $IG\left(k_\eta, S_\eta\right)$ represent the beta and inverse gamma distributions, respectively. Specifically, we set $\mu_0 = 0, \sigma_{\mu 0}^2 = 10, c_1 = 20, c_2 = 1.5,$

Table 3
MCMC estimates of the DC MSV Model for simulated data.

Parameters	True	Mean	NSE	95% interval	INEF.	CD
a_{11}	1.099	1.107	0.043	[0.857, 1.509]	497.5	0.054
a_{21}	−0.330	−0.417	0.027	[−0.736, −0.211]	372.4	0.856
a_{22}	1.099	1.033	0.013	[0.906, 1.182]	290.5	0.544
d	0.8	0.740	0.015	[0.602, 0.866]	385.6	0.304
k	10	8.530	0.486	[5.350, 13.793]	415.1	0.055

The first 2000 draws are discarded and then the next 8000 are used for calculating the posterior means, the standard errors of the posterior means, 95% interval, the inefficiency factor (INEF.) of Kim et al. (1998), and the p-values of the convergence diagnostic (CD) statistics proposed by Geweke (1992). The posterior means are computed by averaging the simulated draws. The numerical standard errors (NSE) of the posterior means are computed using a Parzen window with a bandwidth of 800. The 95% intervals are calculated using the 2.5th and 97.5th percentiles of the simulated draws.

$k_\eta = 5$, and $S_\eta = 0.01 \times k_\eta$. For μ_i, we will report the results of $\sigma_i = \exp(0.5\mu_i/(1-\phi_i))$ instead of μ_i. By conducting the method of Kim et al. (1998) in the first stage, we obtain the standardized series, $z_{it} = \varepsilon_{it} W_{it}$, based on the MCMC draws of the log-volatilities.

In the second stage, we have $z_t \sim N(0, P_t)$ from equations and. Thus, the likelihood function is given by

$$L(A, d, k \mid Z) \propto \prod_{t=1}^{T} \left|P_t^{-1}\right|^{1/2} \exp\left(-\frac{1}{2} z_t' P_t^{-1} z_t\right) \times \frac{\left|S_{t-1}^{-1}\right|^{\frac{k}{2}} \left|Q_{t-1}^{-1}\right|^{\frac{k-m-1}{2}}}{2^{\frac{km}{2}} \Gamma_m(k)} \exp\left(-\frac{1}{2} \mathrm{tr}\left\{S_{t-1}^{-1} Q_t^{-1}\right\}\right)$$

where $S_t = (1/k) Q_t^{-d/2} A Q_t^{-d/2}$ and $\Gamma_m(k) = \prod_{i=1}^{m} \Gamma(k-i+1)$. Again,$\{P_t\}$ are the time-varying correlation matrices obtained by standardizing $\{Q_t\}$. For convenience, we use the fixed value of Q_0^{-1} as $Q_0^{-1} = I_m$.

For the unknown parameters for the process of $\{Q_t\}$, we assume the following prior distributions: $A^{-1} \sim W_m(\gamma_0, C_0)$, $d \sim U(-1, 1)$, and $k \sim \mathrm{EXP}(\lambda_0) I_{(m,\infty)}(k)$, where the distribution for k is the truncated exponential distribution with parameter λ_0. Specifically, we set $C_0^{-1} = \gamma_0 I_m, \gamma_0 = m$,and $\lambda_0 = 5$. As stated above, MCMC sampling requires the conditional posterior distribution for each parameter and Q_t. As the joint posterior distribution is proportional to the prior times the likelihood, the kernel of the conditional posterior distribution is obtained by eliminating irrelevant parameters (see the Appendix for the conditional distributions for (A, d, k) and $\{Q_t\}$, and their sampling methods).

In order to estimate the parameters, (μ, ϕ, σ_η) and (A, d, k), and the unobservable components,$\{h_t\}$ and $\{Q_t\}$, we conduct MCMC simulation with $N + R$ iterations. The first N draws are discarded and then the next R are recorded. Using these R draws for each of the parameters, we calculate the posterior means, the standard errors of the posterior means, the 95% intervals, the inefficiency factor suggested by Kim et al. (1998), p-values, and the convergence diagnostic (CD) statistics proposed by Geweke (1992). The posterior means are computed by averaging the simulated draws. The 'so called' numerical standard error (NSE), which corresponds to Monte Carlo standard errors for numerical integrations, can be used as a measure to check the accuracy of estimated posterior means. The NSEs of the posterior means are computed using a Parzen window with a bandwidth of $0.1R$ (see Geweke (1992) for further details). The 95% intervals are calculated using the 2.5th and 97.5th percentiles of the simulated draws. In order to assess the convergence of the Markov chains, we calculate the CDs, which have the asymptotic standard normal distribution under the null hypothesis of convergence. If the inefficiency factor is one, then each draw of the Markov chain is considered to be a draw from independent sampling. In this case, the Markov chain is ideal in the sense that an independent chain converges to the target distribution quickly. On the other hand, if the inefficiency factor is large, then the Markov chain converges to the target distribution slowly.

3.3. Simulation example

In this subsection, we concentrate on the second stage of the estimation procedure. Data are generated for $\{z_t\}$, with sample size $T = 500$ and true parameters given by:

$$A = \begin{pmatrix} 1 & 0.3 \\ 0.3 & 1 \end{pmatrix}^{-1}, \qquad d = 0.8, \qquad k = 10.$$

In order to estimate these parameters, the MCMC simulation is conducted with 10 000 iterations. The first 2000 draws are discarded and the next 8000 are recorded.

Table 3 shows the results of MCMC estimation. The inefficiency factor is large, indicating that the speed of convergence is slow. The p-values for the CD statistics are greater than 0.05, and hence the Markov chains have converged. Even though the sample size is small for latent variable models, the estimates of the means are close to their true values, and the 95% credible intervals contain the true values.

3.4. Model comparison via simulation

In this section, we compare five models, namely the CCC model of Bollerslev (1990), the DCC model of Engle (2002), the basic MSV model of Harvey et al. (1994), and the CC MSV and DC MSV models, in a setting where the true correlation structure is known. A bivariate SV model is simulated for 1000 observations, or approximately 4 years of daily data, for each correlation process. The processes are given as follows:

$$h_{1,t+1} = 0.98 h_{1t} + \eta_{1t},$$
$$h_{2,t+1} = 0.95 h_{2t} + \eta_{2t},$$
$$\begin{pmatrix} \eta_{1t} \\ \eta_{2t} \end{pmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0.166^2 & 0 \\ 0 & 0.260^2 \end{bmatrix}\right),$$
$$\begin{pmatrix} \nu_{1t} \\ \nu_{2t} \end{pmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & \rho_t \\ \rho_t & 1 \end{bmatrix}\right)$$
$$r_{1t} = \nu_{1t} e^{h_{1t}/2}, \qquad r_{2t} = \nu_{2t} e^{h_{2t}/2}.$$

We use the framework given in Engle (2002) for the correlation models, namely:

1. Constant: $\rho_t = 0.9$;
2. Sine: $\rho_t = 0.5 + 0.4\cos(2\pi t/200)$;
3. Fast Sine: $\rho_t = 0.5 + 0.4\cos(2\pi t/20)$;
4. Step: $\rho_t = 0.9 - 0.5 I(t > 500)$;
5. Ramp: $\rho_t = \mathrm{mod}\,(t/200)$.

Engle (2002) chose these processes as they exhibit various types of rapid changes, gradual changes, and periods of constancy. Some of the processes appear to be mean reverting, while others have abrupt changes. If the processes for the correlation coefficients are non-stochastic, the DCC model has an advantage regarding the estimation of ρ_t, but if the processes for the volatilities are stochastic, the DC MSV model will have an advantage.

The MCMC estimation method, as described above, is used for the parameters and the latent processes of the DC MSV model. The CCC and DCC models are estimated by the quasi-maximum likelihood method, while the Monte Carlo likelihood (MCL) method

$$\mathrm{VaR}_t^0 = 1.65\sqrt{w^2 \exp(h_{1t}) + (1-w)^2 \exp(h_{2t}) + 2w(1-w)\rho_t \exp(0.5h_{1t} + 0.5h_{2t})}.$$

Box I.

Table 4
Mean absolute error of correlation estimates.

Model	CCC	DCC	MSV	CC MSV	DC MSV
Constant	0.0289	0.0376	**0.0112**	0.0122	0.0237
Sine	0.2548	**0.1406**	0.2695	0.2704	0.1711
Fast Sine	0.2552	**0.2208**	0.2641	0.2644	0.2708
Step	0.2500	**0.0891**	0.2500	0.2500	0.1153
Ramp	0.2504	**0.1597**	0.2712	0.2717	0.1646

Table 5
Mean absolute error of value-at-risk (VaR) estimates.

Model	CCC	DCC	MSV	CC MSV	DC MSV
Constant	6.1922	6.3067	0.2973	0.2358	**0.2353**
Sine	3.1841	3.8775	0.2782	0.2771	**0.2239**
Fast Sine	3.8809	4.0370	0.2544	0.2546	**0.2502**
Step	4.1078	4.6154	1.3053	1.7914	**0.2182**
Ramp	3.6418	4.1588	0.3131	0.3128	**0.2183**

of Sandmann and Koopman (1998) and Asai and McAleer (2006) are used for the basic MSV and CC MSV models.

We use two performance measures, both of which are based on mean absolute errors (MAE). The first MAE measure is a simple comparison of the estimated correlations with the true correlations, which is defined as

$$\mathrm{MAE}_\rho = \frac{1}{T}\sum \left|\hat{\rho}_t - \rho_t\right|.$$

The second measure is an evaluation of the estimator for calculating Value-at-Risk (VaR), for which we consider a portfolio with w invested in the first asset and $(1-w)$ in the second. Under normality, the VaR is defined by

$$\mathrm{VaR}_t = 1.65\sqrt{w^2 V_{1t} + (1-w)^2 V_{2t} + 2w(1-w)\hat{\rho}_t V_{1t}^{1/2} V_{2t}^{1/2}},$$

where $V_{it}(i = 1, 2)$ is the estimate of volatility. For example, the MCMC estimates for the volatility is given by averaging $\exp(h_{it}^{(r)})$ for $r = 1, \ldots, R$. Thus, the second MAE measure is given by

$$\mathrm{MAE}_{\mathrm{VaR}} = \frac{1}{T}\sum \left|\mathrm{VaR}_t - \mathrm{VaR}_t^0\right|$$

where VaR_t^0 is given in Box I.

The reported results are based on an equal weighted portfolio with $w = 0.5$.

Table 4 presents the MAE results for the five correlation estimators for five experiments. When the true correlation model is constant, there is little to choose between the five models. In the remaining four cases, the DCC model has the lowest MAE, followed closely by DC MSV in three cases. Although the process of the true correlation coefficient has no innovation term, the DC MSV model is competitive with the DCC model.

Table 5 gives the MAE results for Value-at-Risk (VaR). In all cases the DC MSV model has the smallest MAE, followed by CC MSV and MSV, while the MAE results for DCC are between 15 and 25 times greater than those of DC MSV. These results reflect the differences in the estimators of volatility for the DCC and DC MSV models, as the estimators of the correlation coefficients are very similar. The results for the basic MSV model are close to those of the CC MSV model. Overall, Tables 4 and 5 indicate that the DC MSV model outperforms the remaining models if the correlation coefficients are stochastic.

Table 6
MCMC estimates of the trivariate DC MSV model for Nikkei, Hang Seng and Straits Times.

Parameters	Mean	NSE	95% interval	INEF.	CD
ϕ_1	0.978	0.0004	[0.967, 0.987]	9.698	0.213
$\sigma_{\eta,1}$	0.189	0.0014	[0.153, 0.224]	13.51	0.926
σ_1	1.168	0.0049	[0.989, 1.389]	5.140	0.774
ϕ_2	0.959	0.0007	[0.942, 0.971]	12.37	0.936
$\sigma_{\eta,2}$	0.265	0.0021	[0.213, 0.318]	14.94	0.986
σ_2	1.269	0.0042	[1.126, 1.410]	6.227	0.738
ϕ_3	0.933	0.0009	[0.915, 0.952]	17.28	0.184
$\sigma_{\eta,3}$	0.365	0.0032	[0.314, 0.427]	26.64	0.880
σ_3	0.944	0.0026	[0.857, 1.028]	6.254	0.332

The first 2000 draws are discarded and then the next 4000 are used for calculating the posterior means, the standard errors of the posterior means, 95% interval, the inefficiency factor (INEF.) of Kim et al. (1998), and the p-values of the convergence diagnostic (CD) statistics proposed by Geweke (1992). The posterior means are computed by averaging the simulated draws. The numerical standard errors (NSE) of the posterior means are computed using a Parzen window with a bandwidth of 400. The 95% intervals are calculated using the 2.5th and 97.5th percentiles of the simulated draws.

4. Empirical analysis

In this section, we examine the MCMC estimates of the DC MSV model for three sets of empirical data, namely the Nikkei 225 Index (Nikkei), Hang Seng Index (Hang Seng) and Straits Times Index (Straits Times) returns. The sample period for all three series is 4 January 1986 to 17 July 2002, thereby giving $T = 3773$ observations. Returns R_{it} are defined as $100 \times \{\log P_{it} - \log P_{i,t-1}\}$, where P_{it} is the closing price on day t for stock i. We use the filtered data, $y_{it} = R_{it} - E(R_{it} \mid I_{t-1})$, based on the threshold AR(1) model.

Table 6 gives the MCMC estimates for the first stage. We conducted the MCMC simulation with 6000 iterations. The first 2000 draws were discarded and then the next 4000 were recorded. The inefficiency factors are about ten percent of those listed in Table 3, indicating relatively fast convergence. The p-values for the CD statistics are greater than 0.05, and hence the Markov chains have converged. The NSEs are very small, indicating accuracy of the estimates. The estimates of ϕ_i lie between 0.93 and 0.98, showing high persistence in the log-volatilities. The estimates of $\sigma_{\eta,i}$ lie between 0.18 and 0.37, while the estimates of σ_i are around one. These estimates are typical of empirical univariate SV models.

Based on the MCMC estimates of the log-volatilities in the first stage, we calculated the standardized series, $z_{it} = \varepsilon_{it}\exp\left(-0.5\hat{h}_{it}\right)$. The correlation matrix of z_t is given by

$$P = \begin{pmatrix} 1 & 0.3230 & 0.2751 \\ 0.3230 & 1 & 0.4150 \\ 0.2751 & 0.4150 & 1 \end{pmatrix},$$

indicating that the returns series of Nikkei, Hang Seng and Straits Times are correlated. However, our primary concern lies in the dynamic correlation matrices, P_t, rather than the constant correlation matrix, P, as given above.

Table 7 presents the MCMC estimates for the second stage regarding the bivariate DC MSV models. We conducted the MCMC simulation with 20 000 iterations. The first 10 000 draws were discarded and the next 10 000 were recorded. The p-values for the CD statistics are greater than 0.05, and hence the Markov chains have converged. Although the NSE of k is not small, this is not surprising as the estimated mean itself is not small.

Table 7(a) shows the MCMC estimates for the case of (Nikkei, Hang Seng). The estimate of d is 0.840, and its 95% interval is [0.785, 0.879]. Hence, d is far from zero, indicating a clear presence

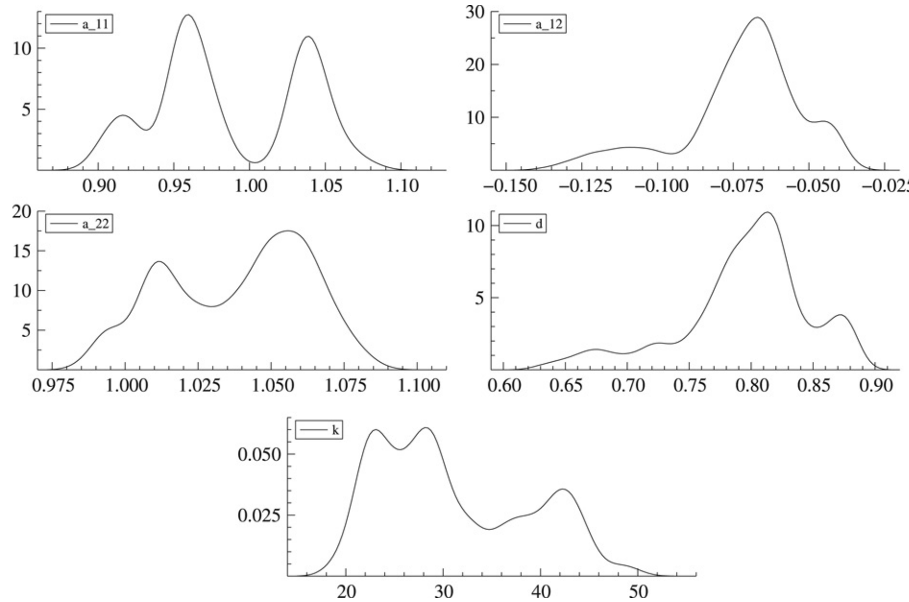

Fig. 1. Posterior densities.

Table 7
MCMC estimates of the bivariate DC MSV model.

Parameters	Mean	NSE	95% interval	INEF.	CD
(a) (Nikkei, Hang Seng)					
a_{11}	1.095	0.005	[1.049, 1.131]	519.4	0.106
a_{21}	−0.059	0.002	[−0.083, −0.041]	473.9	0.151
a_{22}	1.052	0.003	[1.017, 1.080]	504.1	0.306
d	0.840	0.006	[0.785, 0.879]	522.4	0.419
k	31.306	1.164	[23.128, 42.944]	529.9	0.186
(b) (Hang Seng, Straits Times)					
a_{11}	0.991	0.004	[0.964, 1.019]	605.3	0.188
a_{21}	−0.068	0.003	[−0.096, −0.047]	489.8	0.121
a_{22}	1.100	0.006	[1.063, 1.149]	577.8	0.472
d	0.859	0.005	[0.813, 0.897]	509.9	0.076
k	32.188	1.015	[24.854, 41.498]	544.2	0.052
(c) (Nikkei, Straits Times)					
a_{11}	1.010	0.006	[0.971, 1.042]	627.1	0.197
a_{21}	−0.046	0.002	[−0.062, −0.033]	428.7	0.080
a_{22}	1.075	0.005	[1.038, 1.115]	574.9	0.412
d	0.851	0.006	[0.808, 0.903]	555.4	0.149
k	37.079	1.919	[26.280, 54.740]	592.9	0.097

The first 10 000 draws are discarded and then the next 10 000 are used for calculating the posterior means, the standard errors of the posterior means, 95% interval, the inefficiency factor (INEF.) of Kim et al. (1998), and the p-values of the convergence diagnostic (CD) statistics proposed by Geweke (1992). The posterior means are computed by averaging the simulated draws. The numerical standard errors (NSE) of the posterior means are computed using a Parzen window with a bandwidth of 1000. The 95% intervals are calculated using the 2.5th and 97.5th percentiles of the simulated draws.

of dynamic correlations. The estimate of the degrees of freedom parameter, k, for the latent Wishart process is 31.3. For the MCMC estimates of A, all 95 percent intervals exclude zero. As A is the basic matrix for the process of Q_t^{-1}, it is convenient to consider A^{-1} for Q_t and P_t. The estimate of A^{-1} is given by

$$\hat{A}^{-1} = \begin{pmatrix} 0.916 & 0.051 \\ 0.051 & 0.953 \end{pmatrix},$$

implying a weak and positive correlation between the Nikkei and Hang Seng returns.

Fig. 1 shows the estimated densities based on the MCMC draws. The estimated density functions, which are computed using a standard non-parametric Gaussian kernel method, seem multi-modal for all the parameters. Furthermore, the estimated densities are skewed. These points would have to be accommodated if we were to consider the highest posterior density interval.

Figs. 2 and 3 give the estimated conditional and stochastic dynamic correlations between Nikkei and Hang Seng for the DCC and DC MSV models, respectively. For purposes of comparison, Fig. 2 is based on the DCC estimates while Fig. 3 is estimated using the DC MSV model. It should be noted that the scale of the vertical axis in Fig. 2 is two-thirds that of the scale in Fig. 3, with the range of the dynamic conditional correlations in Fig. 2 being (−0.035, 0.598), while the range of the DC MSV dynamic correlations in Fig. 3 is much larger at (−0.242, 0.726). The mean and variance of the dynamic correlations of the DCC estimates are 0.348 and 0.018, respectively. The mean of the DC MSV dynamic correlations is similar at 0.326, but the variance is much higher at 0.029.

Although these figures show that most of the instantaneous correlations are located in the positive region, there was a notable fall around observation $T = 1350$, which is associated with the bursting of the economic and financial bubble in Japan. After appropriate adjustments, the positive correlation between Nikkei and Hang Seng is recovered. In comparison with the DCC model, the DC MSV model is more sensitive to the covariation between the Nikkei and Hang Seng indexes. The relative insensitivity of the DCC estimates is derived from the estimates of the coefficient of $z_{t-1}z'_{t-1}$. As $\hat{\gamma} = 0.0095$ and is close to zero, the correlation structure of the restrictive DCC model does not seem adequate for capturing the movements in $z_{t-1}z'_{t-1}$. Furthermore, $\hat{\delta} = 0.9898$, so that the correlation process for the DCC model has the possibility of a unit root, and hence also constant conditional correlations in the long run In contrast, the estimates arising from the DC MSV model indicate that the dynamic correlation process is stationary.

Table 7(b) and (c) present the MCMC estimates for the bivariate DC MSV model for the pairs (Hang Seng, Straits Times) and (Nikkei, Straits Times), respectively. These are similar to Table 7(a) in that the estimates of d are close to 0.85, while the estimates of k are located in the range [32, 38].

Table 8 shows the MCMC estimates for the trivariate DC MSV model for the three series given by (Nikkei, Hang Seng, Straits Times). The estimates of (a_{21}, a_{31}, a_{32}) are close to the corresponding estimates in Table 7. The estimates of d are close to 0.85, while the 95% interval of d is [0.814, 0.871]. Interestingly, the lower bound is close to the maximum of the three lower bounds for d in Table 7, while the upper bound is the minimum of the three

Fig. 2. Dynamic correlation estimates for bivariate DCC: Nikkei and Hang Seng.

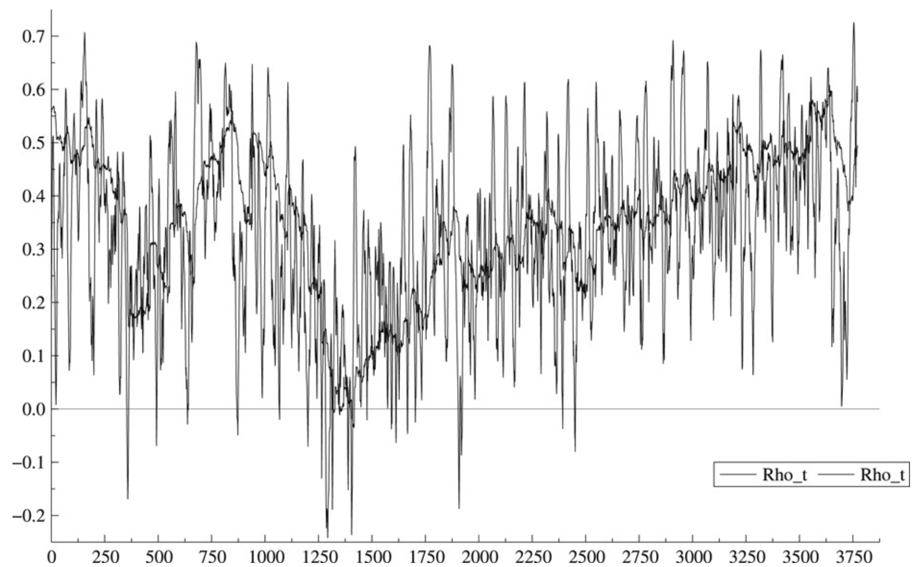

Fig. 3. Dynamic correlation estimates for bivariate DC MSV: Nikkei and Hang Seng.

upper bounds. As the three MCMC estimates of d are close to each other, the 95% interval is shortened by using information for the trivariate data. The estimate of k is close to 39, which is greater than the three estimates shown in Table 7. Upon concentrating three different estimates of d in Table 7 into a single estimate in Table 8, the unexplained movements in the dynamic correlations may be absorbed by the Wishart variable, thereby yielding the larger value of k in Table 8.

Fig. 4 gives the estimated stochastic correlations based on the trivariate DC MSV model. The first panel corresponds to the correlations between Nikkei and Hang Seng, and is reasonably similar to Fig. 3. Compared with Fig. 3, the estimates in Table 4 are less sensitive to shocks, a result that is caused by introducing the third variable in the DC MSV specification. The second panel in Fig. 4 shows the dynamic correlations between Nikkei and Straits Times, while the third panel shows the dynamic correlations between Hang Seng and Straits Times. The variability between Nikkei and Straits Times is greater than those of the other two pair of dynamic correlations.

Table 9 gives the descriptive statistics for the estimated correlations. The means are very close to their constant correlation counterparts that were estimated for P above. While the mean of the dynamic correlations for the pair (Hang Seng, Straits Times) is

Table 8
MCMC estimates of the trivariate DC MSV Model for Nikkei, Hang Seng and Straits times.

Parameters	Mean	NSE	95% interval	INEF.	CD
a_{11}	1.082	0.003	[1.093, 1.105]	530.5	0.176
a_{21}	−0.048	0.001	[−0.063, −0.037]	444.6	0.086
a_{31}	−0.038	0.001	[−0.049, −0.029]	387.2	0.507
a_{22}	1.116	0.004	[1.087, 1.140]	566.5	0.598
a_{32}	−0.070	0.002	[−0.087, −0.056]	495.0	0.069
a_{33}	1.025	0.004	[0.999, 1.055]	657.9	0.421
d	0.846	0.004	[0.814, 0.871]	592.7	0.057
k	39.142	0.970	[31.477, 45.657]	639.4	1.301

The first 10 000 draws are discarded and then the next 10 000 are used for calculating the posterior means, the standard errors of the posterior means, 95% interval, the inefficiency factor (INEF.) of Kim et al. (1998), and the p-values of the convergence diagnostic (CD) statistics proposed by Geweke (1992). The posterior means are computed by averaging the simulated draws. The numerical standard errors (NSE) of the posterior means are computed using a Parzen window with a bandwidth of 1000. The 95% intervals are calculated using the 2.5th and 97.5th percentiles of the simulated draws.

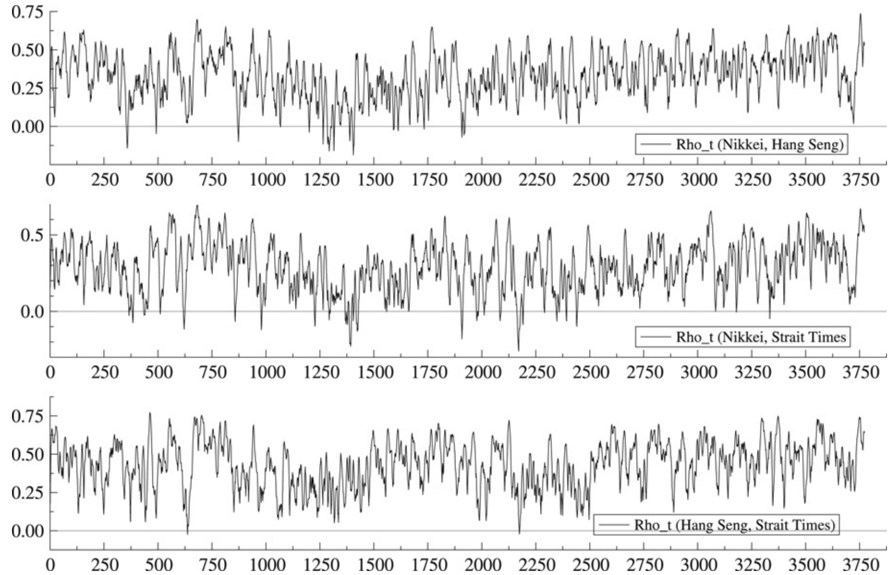

Fig. 4. Pairwise dynamic correlation estimates for trivariate DC MSV.

Table 9
Descriptive statistics for the estimated correlations for Nikkei, Hang Seng and Straits times.

Paired combination	Mean	Var	Min	Max	Correlation	
					(1, 2)	(1, 3)
(Nikkei, Hang Seng)	0.3357	0.0233	−0.1867	0.7369	1	–
(Nikkei, Straits Times)	0.2893	0.0262	−0.2597	0.6945	0.5294	1
(Hang Seng, Straits Times)	0.4346	0.0224	−0.0244	0.7724	0.3995	0.4452

higher at 0.4346 than for the other two paired combinations, the variances are reasonably close to each other, albeit the highest for the (Nikkei, Strat Times) pair. The series (Nikkei, Straits Times) has the widest range, namely (−0.260, 0.695), which ranges from mild hedhing to strong specialization in terms of portfolio selection.

In Table 9, the correlation coefficient between the dynamic correlations of (Nikkei, Hang Seng) and (Hang Seng, Straits Times) has the lowest value at 0.3995, which reflects the relatively weak relationship between the Nikkei and Straits Times stock market indexes. The highest correlation coefficient is between the dynamic correlations of (Nikkei, Hang Seng) and (Nikkei, Straits Times), at 0.5294, which reflects the relatively stronger relationship between the Hang Sengi and Straits Times stock market indexes.

5. Conclusion

As static and dynamic covariance and correlation structures may be used to formulate optimal portfolio choice and risk management strategies, and forecast Value-at-Risk (VaR) threshold estimates, it is essential to model the correlation and covariance matrices accurately. Although the conditional volatility literature has examined the theoretical development of alternative dynamic covariance and correlation structures, this issue does not yet seem to have been examined in detail in the Multivariate Stochastic Volatility (MSV) literature.

This paper has contributed to the development of parsimonious dynamic correlation MSV models that can be estimated with relative ease. Two types of stochastic correlation structures for MSV models were proposed, namely the constant correlation (CC) MSV and dynamic correlation (DC) MSV models. Alternative parsimonious DC MSV models were developed. A technique was developed for estimating the DC MSV model using the Markov Chain Monte Carlo (MCMC) procedure, the properties of the estimation method were examined using simulated data, and various multivariate conditional volatility and MSV models were compared via simulation, including an evaluation of alternative VaR estimators.

The DC MSV model was estimated using three sets of empirical data, namely Nikkei 225 Index, Hang Seng Index and Straits Times Index returns, and significant dynamic correlations were found. The Dynamic Conditional Correlation (DCC) model was also estimated, and was found to be far less sensitive to the covariation in the shocks to the indexes. The correlation process for the DCC model also appeared to have a unit root, and hence constant conditional correlations in the long run. In contrast, the estimates arising from the DC MSV model indicated that the dynamic correlation process was stationary.

Although not considered here, the proposed dynamic correlation models may also be extended to capture leverage effects. Based on the asymmetric MSV models developed in Asai and McAleer (2006), the constant correlation matrix may be replaced by one of several dynamic counterparts. In order to estimate such asymmetric dynamic correlation models, univariate asymmetric SV models, such as those based on the MCMC techniques developed in Omori et al. (2007), would be estimated at the first stage. Such a task awaits future research.

Appendix. Conditional posterior distributions

MCMC methods allow each set of parameters to be sampled separately from its conditional posterior distribution to obtain results that are equivalent to sampling directly from the joint posterior distribution. Within the Gibbs sampler, we sample directly from the posterior distributions of A^{-1}, for the unobservable process $\{Q_t^{-1}\}$, we use the Metropolis algorithm, and for the scalar parameters d and k, we use a flexible sampling approach.

A.1. Conditional posterior distribution of $\{Q_t^{-1}\}$

For the periods $t = 1, 2, \dots, T-1$, the conditional posterior distribution is proportional to the product of the following three terms:

$$\begin{aligned} p\left(Q_t^{-1} \mid \bullet\right) &\propto W_m\left(Q_t^{-1}|k, S_{t-1}\right) \times N\left(0, P_t\right) \times W_m\left(Q_{t+1}^{-1}|k, S_t\right) \\ &\propto \left|Q_t^{-1}\right|^{\frac{k(1-d)-m}{2}} \exp\left(-\frac{1}{2}\mathrm{tr}\left[S_{t-1}^{-1}Q_t^{-1}\right] - \frac{1}{2}z_t'P_t^{-1}z_t \right. \\ &\quad \left. - \frac{1}{2}tr\left[S_t^{-1}Q_{t+1}^{-1}\right]\right) \\ &\propto W_m\left(Q_t^{-1} \mid \hat{k}, \hat{S}_{t-1}\right) \times f\left(Q_t^{-1}\right) \end{aligned} \quad (6)$$

where

$$\hat{k} = k+1, \hat{S}_{t-1} = \left(S_{t-1}^{-1} + z_t z_t\right)^{-1}, S_{t-1} = (1/k)Q_{t-1}^{-d/2}AQ_{t-1}^{-d/2},$$

and

$$\begin{aligned} f\left(Q_t^{-1}\right) = \left|P_t^{-1}\right|^{\frac{1}{2}} \left|Q_t^{-1}\right|^{\frac{-1-kd}{2}} \exp\left(-\frac{1}{2}tr\left[S_t^{-1}Q_{t+1}^{-1}\right]\right. \\ \left. - \frac{1}{2}tr\left[\left(P_t^{-1} - Q_t^{-1}\right)z_t z_t'\right]\right) \end{aligned}$$

is a remainder term. It is assumed that Q_0^{-1} is known. For $t = T$, the conditional posterior distribution is proportional to:

$$p\left(Q_T^{-1} \mid \bullet\right) \propto W_m\left(Q_T^{-1} \mid \hat{k}, \hat{S}_{T-1}\right) \times f\left(Q_T^{-1}\right),$$

where $f(Q_T^{-1}) = \exp\left(-0.5tr\left[\left(P_T^{-1} - Q_T^{-1}\right)z_T z_T'\right]\right)$. The Metropolis algorithm is used to sample from the conditional distribution of $\left\{Q_t^{-1}\right\}$. A candidate $Q_t^{-1[*]}$ is sampled from $W_m\left(Q_t^{-1} \mid \hat{k}, \hat{S}_{t-1}\right)$, where $\hat{S}_{t-1}$ is defined in. The acceptance ratio in the Metropolis algorithm is given by the ratio of the remaining terms, as the candidate density cancels out, $f\left(Q_t^{-1[*]}\right)/f\left(Q_t^{-1[c]}\right)$, where $Q_t^{-1[c]}$ is the current state.

A.2. Conditional posterior distribution of A^{-1}

The matrix A^{-1} is used instead of A as the inverse is more convenient for deriving the sampling distribution. As the conditional distribution of A^{-1} is a product of a Wishart prior and a Wishart likelihood, it also follows a Wishart distribution, namely:

$$\begin{aligned} p\left(A^{-1} \mid \bullet\right) &\propto W_m\left(\gamma_0, C_0\right) \times W_m\left(\gamma, C\right) \\ &\propto \left|A^{-1}\right|^{\frac{\gamma_0-m-1}{2}} \exp\left(-\frac{1}{2}\mathrm{tr}\left\{C_0^{-1}A^{-1}\right\}\right) \\ &\quad \times \left|A^{-1}\right|^{\frac{Tk}{2}} \exp\left(-\frac{1}{2}\mathrm{tr}\left\{C^{-1}A^{-1}\right\}\right) \\ &\propto W_m\left(A^{-1} \mid \hat{\gamma}, \hat{C}\right) \end{aligned} \quad (7)$$

where $\hat{C}^{-1} = C_0^{-1} + C^{-1}, C^{-1} = k\sum_{t=1}^{T} Q_{t-1}^{d/2}Q_t^{-1}Q_{t-1}^{d/2}, \hat{\gamma} = \gamma + \gamma_0 - m - 1 = kT + \gamma_0$, and $\gamma = kT + m + 1$. It is possible to sample directly from the conditional distribution of A^{-1} as a step in the Gibbs sampler.

A.3. Conditional posterior distributions of d and k

The conditional distribution of d is derived from the prior and the terms regarding d, which appear in the likelihood function:

$$\begin{aligned} p\left(d \mid \bullet\right) &\propto p(d)\prod_{t=1}^{T} \left|S_{t-1}\right|^{\frac{-k}{2}} \exp\left(-\frac{1}{2}tr\left\{S_{t-1}^{-1}Q_t^{-1}\right\}\right) \\ &\propto \left|Q_{t-1}^{-1}\right|^{\frac{-dk}{2}} \exp\left(-\frac{1}{2}tr\left\{A^{-1}C^{-1}(d)\right\}\right) I_{(-1,1)}(d) \\ &\propto \exp\left(\psi d - \frac{1}{2}tr\left\{A^{-1}C^{-1}(d)\right\}\right) I_{(-1,1)}(d) \end{aligned} \quad (8)$$

where $\psi = -(k/2)\sum_{t=1}^{T} \ln\left|Q_{t-1}^{-1}\right|$ and $C^{-1}(d) = k\sum_{t=1}^{T} Q_{t-1}^{d/2} Q_t^{-1}Q_{t-1}^{d/2}$ is the same as C^{-1} defined above.

In a similar manner, the conditional posterior distribution of k is given as follows:

$$\begin{aligned} p\left(k \mid \bullet\right) &\propto \exp\left(-\lambda_0 k + \frac{Tkm}{2}\ln(k/2) - \frac{Tk}{2}\ln|A|\right. \\ &\quad \left. - T\sum_{j=1}^{m} \ln\Gamma\left(\frac{k+1-j}{2}\right)\right) \\ &\quad \times \exp\left(\frac{k}{2}\sum_{t=1}^{T}\ln\left|Q_{t-1}^{d/2}Q_t^{-1}Q_{t-1}^{d/2}\right|\right. \\ &\quad \left. - \frac{1}{2}tr\left\{A^{-1}C^{-1}(k)\right\}\right) \end{aligned} \quad (9)$$

where $C^{-1}(k) = k\sum_{t=1}^{T} Q_{t-1}^{d/2}Q_t^{-1}Q_{t-1}^{d/2}$ is the same as C^{-1} defined above.

As the conditional posterior distributions of d and k are complicated, we apply the Adaptive Rejection Metropolis Sampling (ARMS) algorithm of Gilks et al. (1995) in order to sample each draw.

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